Health practices and mortality in Japan: combined effects of smoking, drinking, walking and body mass index in the Miyagi Cohort Study.

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BACKGROUND: Evidence is limited regarding the association between the combinations of multiple health practices and mortality. METHODS: In 1990, 28,333 men and women in Miyagi Prefecture in rural northern Japan (40-64 year of age) completed a self-administered questionnaire. A lifestyle score was calculated by adding the number of high-risk practices (smoking, consuming > or = 22.8 g alcohol/d, walking < 1 hr/d, body mass index < 18.5 or > or = 30.0). Cox regression was used to estimate relative risk (RR) of mortality according to the lifestyle score, with adjustment for age, education, marital status, past history of diseases, and dietary variables. During 11 years of follow-up, 1,200 subjects had died. RESULTS: We observed linear increase in risk of death associated with increasing number of high-risk practices: compared with men who had no high-risk practices, multivariate RRs for men who had 1 to 4 practices were 1.20, 1.66, 1.94, and 3.96, respectively (P for trend<0.001), and corresponding RRs for women were 1.31, 2.14, 3.98, 5.56, respectively (P for trend<0.001). A unit increase in the number of high-risk practices corresponded to being 2.8 and 4.8 years older for men and women, respectively. CONCLUSIONS: In this prospective cohort study of middle-aged men and women in rural Japan, a larger number of high-risk practices was associated with linear increase in risk of all-cause mortality.